

Jillian Rose Brown

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Education

Master of Science in Natural Language Processing

Sep 2025 - June 2027

University of Lorraine | University of Groningen

- Erasmus Mundus Language & Communication Technology Scholar
- Double Degree Program

Bachelor of Arts in Global Studies & Linguistics

Aug 2020 - Dec 2024

South Carolina Honors College, University of South Carolina, *Columbia, SC, USA*

- *Thesis*: “Pragmatic & Discourse Particles of Singaporean English”
- *GPA*: 3.98

Professional Experience

Software Developer, ScholasTech | Wordification

May 2024 - Present

- Implemented voice-cloning and generative speech technology across regional and ethnic accents, managing consent-based voice sample collection and generation for personalized experience
- Built a Python data pipeline that scrapes and structures word-level linguistic data (phonetic transcription, IPA-to-machine-readable conversion, complexity scoring) from open-source resources, creating a reusable dictionary dataset for downstream applications
- Developed dialect- and development-sensitive error detection logic to distinguish dialectal spelling variation from developmental spelling errors, informed by direct research and interviews with educators and literacy specialists
- Contributed to full-stack development (React.js, Redwood.js, PostgreSQL) maintaining and extending a live educational platform

Research Assistant, LORIA Laboratory | University of Lorraine

January - June 2026

- Designed three-stage experimental validation pipeline for evaluation of acoustic indicators of depression in speech through statistically and ML-based methods
- Developed and critically interpreted a binary classifier for depression detection using cross-validated logistic regression, with attention to class imbalance, clinical constraints, and model limitations
- Built a reproducible Python audio preprocessing pipeline integrating multiple third-party toolboxes (librosa, openSMILE, Praat), showcasing practical audio engineering skills for NLP application
- Applied linear mixed-effects models in R to isolate confounding variables in speech feature data

Project Experience

Low-Saxon Dialect Automatic Speech Recognition

Sep 2026 - Present

Designing low-resource dialectal ASR models at University of Groningen, Netherlands

- Designing a transfer-learning evaluation framework to compare fine-tuning strategies for wav2vec2 and Whisper — single-dialect, related-dialect pooling, and high-resource cross-lingual transfer — to identify which training scaffold best supports low-resource dialectal ASR
- Developing a multi-level framework for quantifying dialectal relationships (embedding similarity, phonetic representations, unsupervised clustering) to guide data selection and model scaffolding decisions

Essential Skills

- **Programming & Web Development:** Python, R, JavaScript, CSS/HTML
- **Tools & Frameworks:** PyTorch, SQL, React, Git, Audacity, Praat
- **Skills:** Statistical and ML Model Evaluation, Fine-Tuning, AI Agenrts, Speech Model Systems, Retrieval Augmented Generation, App Development, Prompt Engineering, Large Data Sets